

In T.306/10 the board cited T.1741/08 and decided that the selection of an item for recommendation to a user did not qualify as a technical purpose. From a technical point of view the recommendations were irrelevant, because while making "good" or "bad" recommendations might lead to different user reactions and thereby, in the end, to different technical results, such results did not qualify as a technical effect of the recommendations, as they depended on subjective choices made by the user.

In T.1834/10 a web page is made more attractive to potential customers by presenting images in a lively manner, and the images to be presented are selected and displayed automatically so that the presentation can be changed rapidly. The board noted that however innovative an attractive content selection may be, it related to a presentation of information which was a priori non-technical (Art. 52(2)(d) EPC), even if it **lowers a user's cognitive burden** (T.1741/08) or prompts the user to start some technical action (**broken technical chain**, T.1741/08, T.1670/07). The desire to present a selection of images, to limit the selection to a predetermined number of images and to present the selected images in some **preferred layout** is a non-technical aim. The meaning of the display data does not convey any technical character to the presentation.

The board in T.49/04 decided otherwise, finding that technical aspects could not be ruled out in the design and use of a graphic interface. Furthermore, the presentation of natural language text on a display in a manner which improved readability, enabling users to perform their task more efficiently, related to how, i.e. by what physical **arrangement of the text**, cognitive content was conveyed to the reader and could thus be considered as contributing a technical solution to a technical problem.

In T.619/98 it was held that an action performed by a user in response to a message in the form of questions or suggestions concerning the technical functioning of an apparatus did not render the form of the information technical. The board in T.1143/06 came to the same conclusion, holding that an action (possibly) performed by a user in response to a message concerning the technical functioning of an apparatus did not necessarily render technical the information conveyed. The board came to the conclusion that the information provided according to the relevant features could not credibly support a **continued and guided human-machine interaction** process. Thus, it could not assist the user in performing the technical task. It only addressed the human mental process of an "average user", however the latter was supposed to be defined based on **personal skills and preferences** (see e.g. T.407/11).

ii) Features of the graphic design

In T.244/00 the board stated that the **graphic design of menus** was, as a rule, not a technical aspect of a menu-driven control system. Nor was the practical use of such menus genuinely a problem with which the skilled person, in his function as a technical expert, was confronted. For the purpose of the problem and solution approach, the problem had to be a technical problem which the skilled person in the particular technical field might have been asked to solve at the priority date. The board therefore concluded that, in the case at issue, the technical problem had to be formulated in a more limited way than on

the basis of the alleged advantages of moving a cursor diagonally over the TV screen (see T.154/04, OJ 2008, 46; T.125/04, T.1143/06).

T.333/95 concerned using a computer to produce animations. In order to produce a scene in which a particular object moves, the selected object takes the place of the cursor, so that the user can move it around with the mouse as desired, while the system records the movements of the mouse and translates them into a script of commands for moving the object in the later display of the animated scene. However, the board in that case clearly considered the feature of "making said graphics object the current cursor" as a technical feature per se. This "graphics object cursor" replaced the normal cursor (also technical) and its movements were recorded and translated into a kind of programming language. There was at least a technical effect in allowing the user to move the selected object around under control of the mouse.

In T.125/04 the board stated that, in general, the task of **designing diagrams** is non-technical. This is so even if the diagrams arguably convey information in way which a viewer may intuitively regard as particularly appealing, lucid or logical. In T.125/04 the invention, like T.643/00, also concerned an **arrangement of images** but was different in that only the information conveyed by the images, i.e. their "cognitive content" (see T.1194/97, OJ 2000, 525), was relevant. The new features had to do with how this content was represented. Unlike T.643/00, the invention provided no information about the computer system itself, such as the location where the data are stored.

In T.1073/06, the underlying GUI was configured, upon user input, to display objects of a **simulation model**, including graphical link representations to improve the ease of a user's comprehension of the model. The contribution of the claimed subject-matter to the prior art was related to the utilisation of association data stored in the memory to cause the link between the objects in the simulation model to be displayed with the associated graphical link representation. The deciding board held that "an improvement in the comprehension of a model is a purely mental effect, so that the problem solved was not seen as being technical. The claimed 'graphical link representations' related to the state of the simulation model, rather than to the state of the claimed simulation apparatus, and thus constituted presentations of information and are therefore also non-technical" (see T.336/14).

iii) Support in performing a technical task

In T.1741/08 the board held that not everything that supports a technical task has itself a technical character.

In T.1802/13 the issue was "how" specific cognitive data was actually presented rather than "what" was presented. The crucial consideration was therefore that the feature involved presenting cognitive content which addressed solely the user's mental process. It was established case law of the boards of appeal that, as a general rule, "lowering the cognitive burden of a user" could not, per se, be considered to be a technical effect (see e.g. T.1741/08, T.1539/09 and T.1237/10).

In T 336/14 the board held that in the assessment of inventive step of a claim, in which the non-technical features relate to cognitive content presented to the user of a graphical user interface, i.e. relate to "what" is presented rather than "how" something is presented, it has to be analysed whether the GUI together with the content presented credibly assists the user in performing a technical task (related to "why" that content is presented) by means of a continued and/or guided human-machine interaction process. It has to be established whether the information supplied based on the relevant features credibly enable the user to properly operate the technical system in question, or whether it only addresses the human mental process of an "average user" (on this point, see also T 1895/17 and T 772/18).

In T 690/11 the invention concerned a dialysis system which comprised a display device, a web server and web browser operating with the display device to display information that could guide an operator through the setup procedure for performing a dialysis treatment and then illustrate the progress of that treatment. In the board's opinion, the claimed features possessed more than mere information content directed exclusively to the human mind. The claimed display related to the interaction between the system and the operator and hence, implied technical means for the transmission and handling of respective signals contributing to the correct operation of the system. This conferred a technical character on the claimed features which had thus to be considered in the assessment of novelty and inventive step.

c) Functional and cognitive data

In T 1194/97 (OJ 2000, 525) the board held that functional data were to be distinguished from coded cognitive content. A record carrier characterised by having functional data recorded on it was not a presentation of information as such. In this context the term "functional data" included a data structure defined in terms which inherently comprised the technical features of the system in which the record carrier was used.

In T 2049/12 the board held that a common misconception regarding decision T 1194/97 was that there were only two kinds of data – cognitive and functional – and that functional (i.e. non-cognitive) data was always technical. The relevant question for assessing whether a data structure had technical character was rather whether it produces a technical effect.

In T 643/00 the board noted that it was true that non-technical aspects may be found in the design and the use of an interface through which the user interacts with a system (see decision T 244/00). Indeed, presenting information through a user interface, if the only relevant effect of the presentation relates to the visually attractive nature of the graphic design or artwork, does not have technical character. However, in its decision the board had not excluded the possibility that an arrangement of menu items (or images) on a screen might be determined by technical considerations. Such considerations might be intended to enable the user to manage a technical task, such as **searching and retrieving images** stored in an image processing apparatus, in a more efficient or faster manner, even if an evaluation by the user on a mental level was involved. Although such evaluation per se did not fall within the meaning of "invention" pursuant to Art. 52 EPC 1973, the **mere**

fact that mental activities were involved did not necessarily qualify subject-matter as non-technical. The board referred to T.1177/97 where it was stated that the use of a piece of information in a technical system, or its usability for this purpose, may confer a technical character on the information itself in that it reflects the properties of the technical system, for instance by being specifically formatted or processed.

T.677/09 was concerned with information about differences between different vehicles which was provided to the user of a vehicle information system on actuating a component. The board considered that such an effect would depend on the content of the information and the user's reaction to it. This effect was thus not the **direct effect of the feature** and could not be used to formulate the technical problem. The board noted that the appellant invoked non-technical aspects as a reason for not modifying the prior art and that in T.1670/07, this kind of argument was referred to as a **"non-technical prejudice fallacy"** (point 16 of the Reasons). The fact that this was not possible followed from the fact that the non-technical features relating to the information content could not contribute towards an inventive step at all. The question was not whether the skilled person would consider providing these features because that had already been decided in formulating the technical problem, but whether it would be obvious to implement the features in the claimed manner. The appellant pointed out that in data processing a signal generally has cognitive information content, but according to the jurisprudence nevertheless has technical character. However, the board considered that this technical character was due to the so called "functional data" implied by the signal, which inherently comprises technical features that interact with those of the system in which the signal is operating, such as synchronising data (see for example T.1194/97, point 3.3 of the Reasons). In the case at issue, there were no such inherent technical features of the different information so that it did not have a functional part, but remained purely cognitive. This was different in the case of systems with features that might have a **direct technical effect**, such as giving information about the status of the engine, or about an imminent collision, or how to park the vehicle.

d) User preferences for displaying data

In T.756/06 the board noted that the indication of internal states of a technical system, in the form of visual feedback for human interaction with the system, has been acknowledged to be technical by the boards in the past (see e.g. T.115/85 (OJ 1990, 30; T.362/90). This finding has been confirmed by more recent cases, in particular in T.643/00, where the design of a GUI (Graphical user interface) was seen in the context of the technical process of fast and efficient image retrieval in an image processing apparatus. On the other hand, in cases where the GUI design aimed exclusively at the mental activities of a viewer, in particular at preparing the relevant data for a non-technical decision making process by the user as the final addressee, no technical contribution has been acknowledged beyond its mere implementation. In the case before it, the board considered that the **layout** of the schedule was for the user's mental use, i.e. according to the **user's requirements and preferences**, rather than for a technical purpose in a technical process. Although the appellant mentioned the possibility of user input of the first time span, the board considered that this only resulted in a subjective **improvement** of the appearance of the schedule and was not a part of any technical process.

In T. 1841/06 the board stated that the aim and object of the invention was at best the result of balancing various **mental preferences of the user** but it was per se not a technical problem. Having the option of choosing between an original language and the preferred language might be felt as an inconvenience by one user but as an advantage by another. The invention brought about a mental simplification and subjective advantage for some users but it did not provide any objective advantage nor any technical advance in any field of technology.

In T. 478/06 the claimed invention was a method of providing access to geographic information in a computer system. The board noted that, generally, any aspects that are based on the subjective interests, personal preferences and (business/commercial) activities or circumstances of the user are non-technical in nature. Similarly, managing, i.e. using depending on wishes, personal and geographical information was non-technical. Also, the choice of where to provide a control button was a matter of user preference, and/or the commercial circumstances such as which program was available to be modified. The same goes for the second aspect of prompting the user, when the map information was displayed, to indicate if directions were required, neither being a technical consideration.

In T. 1000/09 the board held that effects resulting from a user-defined data presentation depended on the user's perception and/or constituted **indirect technical effects** and/or related to organisational and economic aspects. Regarding the technical, inputting side of the man-machine interface, the desire to provide it with inputting means for controlling the data output was driven by the obvious needs of users.

In T. 862/10 held that choosing the location of the display object in function of the urgency of the message was non-technical. The board judged that determining (or attempting to determine) a user's visual focus of attention as a point on a screen and displaying objects in positions relative to that point could be considered to have a technical effect, but that the particular choice of where to display an object dependent on a value assigned to that object (its "urgency") could not be.

In T. 1472/14 the board found that what the claimed subject-matter did was merely to organise anthropometric data in a database so that they were provided in standardised form or in the form of statistical characteristic values for querying by way of a communication device. The subject-matter of the claim was concerned with evaluation results only, even if those results were transmitted for a stated purpose of producing products. There was no monitoring of the operation of a production plant; instead, only product data were provided. The board did not consider this to be a technical effect going beyond the mere and obvious automation of an abstract idea for standardisation.

e) Data structures

Data structures which are used to store cognitive data are not considered to contribute to the technical character beyond the mere storage of data, but data structures used for functional purposes are considered to contribute to producing a technical effect (e.g., T. 1194/97, OJ 2000, 525; T. 424/03; T. 697/17).